STABILITY / UPTIME

⚠ CRITICAL

SILENT DATA CORRUPTION

Write operations have been silently corrupting records for 48 hours. Nobody noticed until a paying customer complained that their historical data had vanished. Forensics are underway — but the full scope is still unknown.

Silent data corruption is particularly dangerous because it compounds over time before detection, potentially invalidating large volumes of data and triggering regulatory obligations. Prevention requires multiple layers: write-path validation (schema enforcement, constraint checks, checksums on critical fields), read-path anomaly detection (alerting on unexpected null rates, outlier values, or referential integrity failures), regular data quality audits, and event sourcing or write-ahead logging so every mutation is traceable. Comprehensive observability extends beyond service metrics to data health metrics. The DORA research shows that elite teams have significantly shorter mean time to detect (MTTD) — silent failures are a MTTD problem at their core.

Silent failures are the most dangerous kind. They erode trust long before anyone sounds an alarm.

TEAM DISCUSSION

Do you have checksums, data validation, and anomaly detection on writes? How would you even know this was happening?